Computer Network And Security

Unit I: APPLICATION LAYER

- **Client-Server Paradigm**
  - Communication using TCP and UDP
- **Peer-to-Peer Paradigm**
- **Application Layer Protocols:**
  - DNS (Domain Name System)
  - FTP (File Transfer Protocol)
  - TFTP (Trivial File Transfer Protocol)
  - HTTP (Hypertext Transfer Protocol)
  - SMTP (Simple Mail Transfer Protocol)
  - POP (Post Office Protocol)
  - IMAP (Internet Message Access Protocol)
  - MIME (Multipurpose Internet Mail Extensions)
  - DHCP (Dynamic Host Configuration Protocol)
  - TELNET

Application Layer

1. Definition and Role

The Application Layer is the topmost layer (Layer 7) in the OSI model and the highest layer in the TCP/IP suite. It does not provide services to any other layer; instead, it provides services directly to the software applications (like Chrome, Outlook, or FileZilla) that want to communicate over a network.

2. Key Concepts (As per Unit I)

To score well, you must explain the two primary communication paradigms:

- **Client-Server Paradigm:** A centralized model where a dedicated server provides resources or services, and clients request them (e.g., browsing a website via HTTP).

- **Peer-to-Peer (P2P) Paradigm:** A decentralized model where every node (peer) can act as both a client and a server, sharing resources directly without a central server (e.g., BitTorrent).

3. Communication using TCP/UDP

Explain that the Application Layer relies on the Transport Layer for data delivery:

- TCP (Transmission Control Protocol): Used when reliability is critical (e.g., HTTP, FTP, SMTP). It ensures data is error-free and in the correct order.
- UDP (User Datagram Protocol): Used when speed is more important than perfect reliability (e.g., DNS, TFTP, or Video Streaming).

4. Essential Protocols

Briefly define the specific protocols listed in your syllabus:

Protocol	Full Name	Primary Purpose
DNS	Domain Name System	Translates domain names (google.com) to IP addresses.
HTTP	Hypertext Transfer Protocol	Used for accessing and transferring web pages.
FTP / TFTP	File Transfer Protocol	Used for transferring files between a client and server.
SMTP	Simple Mail Transfer Protocol	Used for sending emails.
POP / IMAP	Post Office / Internet Message Access	Used for receiving/retrieving emails.

DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP addresses to devices on a network.
TELNET	Teletype Network	Provides a bidirectional interactive text-oriented communication facility (Remote Login).

5. Main Functions

Summarize the layer's responsibilities:

- Network Virtual Terminal: Allowing a user to log on to a remote host.
- File Transfer, Access, and Management (FTAM): Moving and managing files.
- Mail Services: Providing the basis for email forwarding and storage.
- Directory Services: Providing access to distributed database sources for global information about various objects and services.

#Client-Server Paradigm

1. Core Concept

In this model, the network roles are clearly divided into two specific entities:

- Server: A powerful, "always-on" host that provides resources or services. It sits in a passive state, waiting for requests.
- Client: A local machine (like your laptop or phone) that requests services from the server.

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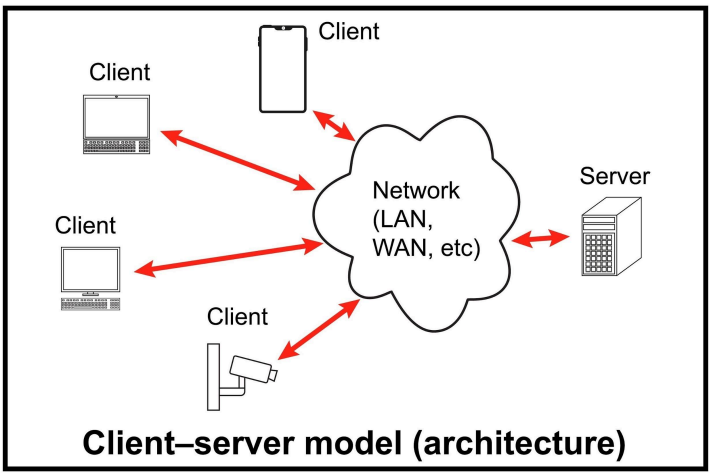
2. Key Characteristics

- **Asymmetric Communication:** The relationship is not equal. The client always starts the communication, and the server responds.
- **Centralization**: Resources (like website files or databases) are stored centrally on the server, making them easier to manage and secure.
- **Scalability**: You can add many clients to a single server, though the server may eventually become a bottleneck.
- **Static IP**: The server usually has a fixed (static) IP address so that clients always know where to find it.



Client–server model (architecture)

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3. The Communication Process

The interaction follows a specific request-response cycle:

1. Request: The client sends a request over the network (e.g., "Give me index.html").
2. Processing: The server receives the request, verifies permissions, and retrieves the data.
3. Response: The server sends the requested data or an error message back to the client.

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4. Examples (Protocols)

Most Application Layer protocols mentioned in your syllabus follow this paradigm:

- HTTP: Your browser (client) requests a page from a web server.
- FTP: A client uploads or downloads files from a central file server.
- DNS: A client asks a DNS server to resolve a domain name into an IP address.
- SMTP/POP: A mail client sends or retrieves emails from a mail server.

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5. Advantages and Disadvantages

Advantages	Disadvantages
Centralized Control: Easy to update data and backup files in one place.	Single Point of Failure: If the server goes down, no client can access the service.

Security: Access can be strictly controlled at the server level.	Congestion: If too many clients request at once, the server slows down (bottleneck).

Communication using TCP and UDP

In the Application Layer, protocols choose their "transport vehicle" based on the needs of the data being sent.

A. Communication using TCP (Transmission Control Protocol)

TCP is used when **reliability** is the top priority.

- **Connection-Oriented:** A "handshake" is established before data is sent.
- **Error Control:** It ensures that every packet arrives and is in the correct order. If a packet is lost, it is retransmitted.
- **Flow Control:** Prevents the sender from overwhelming the receiver with too much data.
- **Examples:** HTTP (Web), FTP (File Transfer), SMTP (Email).

B. Communication using UDP (User Datagram Protocol)

UDP is used when **speed and efficiency** are more important than 100% accuracy.

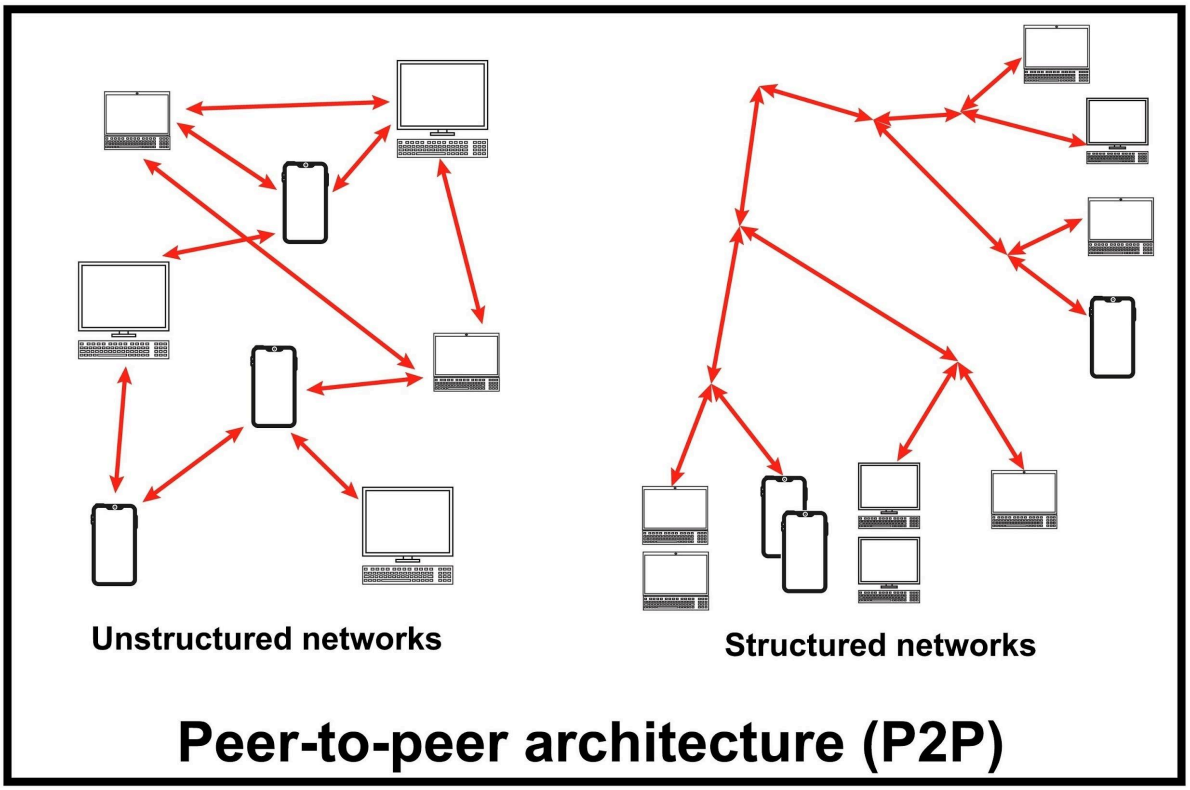
- **Connectionless:** Data is sent without verifying if the receiver is ready.
- **Best-Effort Delivery:** There is no guarantee that packets will arrive or stay in order. There is no retransmission of lost data.
- **Low Overhead:** No handshake or flow control means data moves faster and uses less bandwidth.
- **Examples:** DNS (Domain lookup), Video Streaming, VoIP (Internet calls), TFTP.

Peer-to-Peer (P2P)

1. Definition and Concept

Unlike the Client-Server model, the P2P paradigm is a **decentralized** architecture. In this model, there is no fixed division between clients and servers. Every node (computer) in the network is called a **"Peer."**

- **Self-Scalability:** As more peers join the network, the overall capacity of the system increases because each new peer adds service capacity (bandwidth, storage, and CPU) to the network.
- **Equivalency:** Each peer acts as both a **client** (requesting data) and a **server** (providing data) simultaneously.



2. Key Characteristics

- **No Central Authority:** There is no central server managing the network or storing all the data.
- **Dynamic Connectivity:** Peers are typically connected to the internet with high-churn rates; they join and leave the network frequently and may change their IP addresses.
- **Direct Communication:** Peers communicate directly with each other to share resources (files, processing power, or streaming data).

3. Advantages (Why use P2P?)

- **Cost-Effective:** There is no need for expensive, high-capacity central servers or dedicated administrators.
- **High Reliability/Fault Tolerance:** Since there is no central "point of failure," the network continues to function even if several peers go offline.
- **Scalability:** The network handles more load naturally as it grows; more users mean more sources for the data.

4. Challenges (Disadvantages)

- **Security:** It is harder to manage security and prevent the spread of malware since there is no central gatekeeper.
- **Management:** It is difficult to backup data or ensure data is always available if the specific peer holding that data goes offline.
- **ISP Issues:** P2P traffic can be very heavy and is often throttled by Internet Service Providers.

5. Examples of P2P Applications

- **File Sharing:** BitTorrent, Gnutella.
- **Communication:** Early versions of Skype (using "Supernodes").
- **Blockchain/Crypto:** Bitcoin and Ethereum utilize P2P networks to maintain ledgers.

Comparison: Client-Server vs. P2P

Feature	**Client-Server**	**Peer-to-Peer (P2P)**
Control	Centralized	Decentralized
Cost	Expensive (Server hardware)	Low cost (Uses existing PCs)
Performance	Drops as more clients join	Improves as more peers join
Stability	Depends on the central server	Highly robust

Introduction to DNS

1. DNS

The **Domain Name System (DNS)** is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet.

- **Primary Purpose:** It translates human-readable domain names (e.g., `www.google.com`) into machine-readable IP addresses (e.g., `192.168.1.1`).
- **Protocol/Port:** It primarily uses **UDP** on **Port 53** for queries (due to speed) and TCP for zone transfers.
- **Analogy:** It acts as the "Phonebook of the Internet."

2. The Hierarchical Name Space

DNS uses a tree-like inverted structure called the **Domain Name Space**.

- **Root Level:** Represented by a dot (`.`). There are 13 logical root servers worldwide.
- **Top-Level Domains (TLD):** Divided into two types:
  - **Generic TLDs (gTLD):** `.com`, `.org`, `.edu`, `.net`.
  - **Country Code TLDs (ccTLD):** `.in` (India), `.uk` (United Kingdom).
- **Second-Level Domains:** These are names registered by individuals or organizations (e.g., `google` in `google.com`).
- **Subdomains:** Smaller divisions created by the owner (e.g., `mail.google.com`).

3. Components of DNS

1. **Resolver:** The client-side software (usually in your OS or Browser) that initiates the DNS request.
2. **Name Servers:** Servers that store the database of domain-to-IP mappings.
   - **Recursive Resolver:** The "middleman" (usually your ISP) that hunts down the IP for you.
   - **Authoritative Server:** The final server that holds the actual record for the domain.

4. The DNS Resolution Process (Step-by-Step)

Explain how a query is resolved when a user types a URL:

1. **Request:** The Resolver checks the local cache. If not found, it asks the **Recursive DNS Server**.
2. **Root Server:** The Recursive server asks the **Root Server** for the TLD.
3. **TLD Server:** The Root points to the `.com` TLD server.
4. **Authoritative Server:** The TLD server points to the **Authoritative Name Server** of the specific domain.
5. **IP Delivery:** The Authoritative server provides the IP address back to the Recursive server.
6. **Caching:** The Recursive server gives the IP to the client and caches it for future use.

5. Types of DNS Queries

- **Recursive Query:** The client demands the server to provide the final IP address or an error message (The server does all the "hunting").
- **Iterative Query:** The server provides the best answer it has or refers the client to a different server (The client does the "hunting").

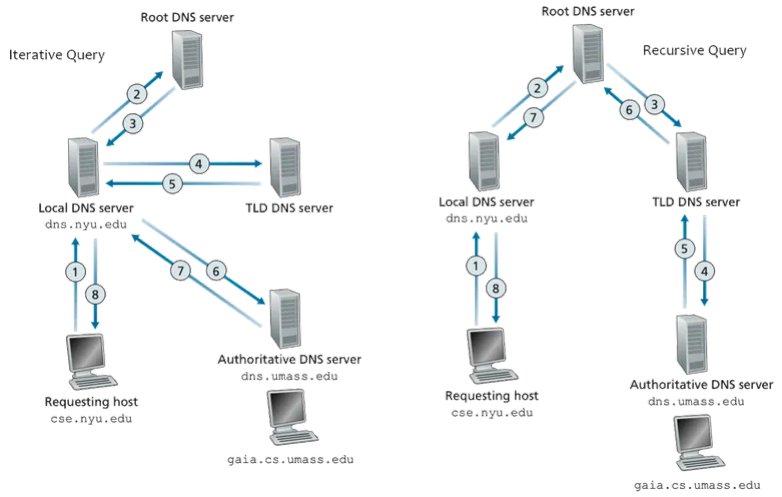
6. Common DNS Resource Records (RR)

A DNS database consists of records. Mention these four:

- **A Record:** Maps a hostname to an IPv4 address.
- **AAAA Record:** Maps a hostname to an IPv6 address.
- **CNAME (Canonical Name):** Maps an alias to the true domain name (e.g., `web.site.com` to `site.com`).
- **MX (Mail Exchanger):** Identifies the mail server for the domain (used by SMTP).

7. Importance of DNS

- **Efficiency:** Users don't need to remember complex numbers.
- **Flexibility:** If a website moves to a new server (new IP), the admin only needs to update the DNS record; the URL remains the same for users.



FTP (File Transfer Protocol)

1. Introduction to FTP

- **Purpose**: It is used for the reliable and efficient transfer of files (uploading and downloading) between a local and a remote host.
- **Model**: It follows the Client-Server Paradigm.
- **Protocol/Port**: It uses TCP for reliable delivery and utilizes two different ports: 21 and 20.

2. Two-Connection Architecture

Unlike most protocols that use a single connection, FTP establishes two parallel connections between the client and the server:

1. Control Connection (Port 21):
    - This connection remains open for the entire duration of the FTP session.
    - It is used for sending "control information" like user IDs, passwords, and commands (e.g., PUT, GET, LIST).
    - It is called Out-of-Band signaling because the control data is separate from the actual file data.
2. Data Connection (Port 20):
    - This connection is opened and closed for each file transfer.
    - It is used specifically for the transfer of the actual file bits.
    - Once the file transfer is complete, this connection is terminated.

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3. FTP Communication Modes

You should mention how the data connection is established:

- Active Mode: The client tells the server its IP and port, and the server initiates the data connection to the client. (Often blocked by firewalls).
- Passive Mode (PASV): The server tells the client which port it is listening on, and the client initiates the data connection. This is the modern standard used to bypass firewalls.

4. Key Features

- Interactive Access: Users can browse directories and delete/rename files on the remote server.
- Authentication: Requires a username and password (though "Anonymous FTP" allows access without an account).
- Data Representation: Can transfer files in ASCII (text) or Binary (images, executables) formats.

5. TFTP (Trivial File Transfer Protocol) - Quick Comparison

Since your syllabus mentions TFTP, it is good to add a small comparison:

Feature	FTP	TFTP
Transport Protocol	TCP (Reliable)	UDP (Best-effort)
Port	21 & 20	69
Security	Supports Authentication	No Authentication
Complexity	Complex, many features	Simple, small memory footprint
Usage	General file sharing	Booting diskless workstations

TFTP

1. Introduction to TFTP

Trivial File Transfer Protocol (TFTP) is a simple, high-level protocol used for transferring files between a client and a server.

- **Design Goal:** It was designed to be small and easy to implement, making it ideal for devices with limited memory or processing power.
- **Transport Layer**: It uses UDP (User Datagram Protocol) on Port 69.
- **Reliability**: Since UDP is unreliable, TFTP must implement its own basic error-checking and acknowledgement system.

2. Key Characteristics

- **No Authentication**: Unlike FTP, TFTP does not support usernames or passwords. It assumes the network is secure or the device is authorized.
- **Single Connection**: It uses one connection for both commands and data (unlike FTP's dual-port system).
- **Fixed Block Size:** Data is sent in fixed-sized blocks of 512 bytes.
- **Stop-and-Wait Mechanism**: It sends one block and waits for an Acknowledgement (ACK) before sending the next one.

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3. TFTP Message Types

TFTP uses five types of packets to manage communication:

1. **Read Request (RRQ):** Client requests to read (download) a file from the server.
2. **Write Request (WRQ)**: Client requests to write (upload) a file to the server.
3. **Data (DATA):** Contains the actual file content (512 bytes per block).
4. **Acknowledgment (ACK)**: Confirms that a data block or request was received.
5. **Error (ERROR)**: Sent when a file isn't found, the disk is full, or an illegal operation occurs.

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4. How TFTP Works (The Process)

1. **Initialization:** The client sends an RRQ or WRQ to the server on Port 69.
2. **Connection:** The server opens a temporary port to communicate with the client.
3. **Data Transfer:** * The sender sends a DATA packet (Block #1).
    - The receiver sends an ACK (Block #1).
    - This continues until a DATA packet is sent that contains less than 512 bytes, signaling the end of the file.

4. **Error Handling:** If a packet is lost, the sender waits for a "Timeout" and retransmits the last block.

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5. Comparison: FTP vs. TFTP

This table is essential for a 10-mark answer to show depth of knowledge:

Feature	FTP	TFTP
Protocol	TCP (Connection-oriented)	UDP (Connectionless)
Ports	20 (Data) and 21 (Control)	69 (Single port)
Security	Requires Login/Password	No Security/Authentication
Complexity	High (Many commands)	Low (Only 5 message types)
File Listing	Can list/browse directories	Cannot list directories
Main Use	General file sharing	Booting diskless workstations

HTTP (Hypertext Transfer Protocol)

HTTP is the foundation of data communication for the World Wide Web.

- **Definition**: An application-layer protocol for transmitting hypermedia documents, such as HTML.
- **Protocol**/Port: Uses TCP (for reliability) on Port 80 (HTTPS uses 443).
- **Stateless Protocol**: Each request is independent; the server does not "remember" previous requests (Cookies/Sessions are used to bypass this).

Key Components:

1. **Request-Response Model: The client (browser) sends a request, and the server (web server) sends a response.**
2. **HTTP Methods:**
    - **GET: Request data from a resource.**
    - **POST: Submit data to be processed (e.g., login forms).**
    - **PUT: Update a resource.**
    - **DELETE: Remove a resource.**

Message Structure:

- **Request Message:** Request Line (Method, URL, Version), Headers, and an optional Body.
- **Response Message:** Status Line (Version, Status Code, Phrase), Headers, and the Entity Body (the actual HTML/Image).

Status Codes to Mention:

- **200 OK: Success.**
- **301 Moved Permanently: Redirect.**
- **404 Not Found: Client error (page doesn't exist).**
- **500 Internal Server Error: Server-side failure.**

SMTP (Simple Mail Transfer Protocol)

SMTP is the standard protocol for sending emails across the Internet.

- **Role**: Acts as an MTA (Mail Transfer Agent) to push mail from the client to the server or between servers.
- **Protocol**/Port: Uses TCP on Port 25 (Standard) or Port 587 (Secure/Submission).
- **Communication**: Uses ASCII commands to communicate between the sender and receiver.

The SMTP Process:

1. **Connection Establishment: Client establishes a TCP connection to the server.**
2. **Mail Transfer:**
    - **HELO: Client identifies itself.**
    - **MAIL FROM: Identifies the sender's address.**
    - **RCPT TO: Identifies the recipient.**
    - **DATA: The actual content of the email is sent here.**
3. **Connection Termination: QUIT command ends the session.**

POP3 (Post Office Protocol version 3)

POP3 is a simple protocol used by local email clients to retrieve emails from a remote server.

- **Concept: Based on the "Store-and-Forward" or "Download-and-Delete" model.**
- **Protocol/Port: Uses TCP on Port 110.**

Phases of POP3:

1. **Authorization**: The user provides a username and password.
2. **Transaction**: The client retrieves the list of emails, downloads them, and marks them for deletion from the server.
3. **Update**: Once the client quits, the server permanently deletes the downloaded emails.

Advantages/Disadvantages:

- **Pros**: Ideal for users with slow or intermittent internet; mail can be read offline.
- **Cons**: Emails are only available on one device. If the device crashes, the emails are lost because they were deleted from the server.

IMAP (Internet Message Access Protocol)

IMAP is a more advanced protocol for retrieving and managing emails.

- **Concept**: Based on the "Download-and-Keep" or "Sync" model.
- **Protocol**/Port: Uses TCP on Port 143.
- **Functionality**: Allows users to view and organize emails directly on the server without downloading the entire message first.

Key Features (Why it's better than POP3):

1. **Multiple Device Sync**: Since mail stays on the server, you can see the same "Sent" and "Drafts" folders on your phone, laptop, and tablet.
2. **Partial Downloa**d: You can download just the email headers or the text and only download heavy attachments if you choose to.
3. **Server-Side Search**: You can search through thousands of emails using the server's processing power rather than your phone's.
4. **Folder Management**: Users can create, delete, or rename folders (mailboxes) directly on the server.

Comparison Table for Exam:

Feature	**POP3**	**IMAP**
Storage	**Downloads to local device.**	**Stays on the server.**
Deletion	**Deleted from server after download.**	**Remains on server until user deletes it.**
Multi-Device	**Not supported (Single device).**	**Fully synchronized across devices.**
Bandwidth	**Saves bandwidth after first download.**	**Uses more bandwidth (Constant syncing).**

MIME (Multipurpose Internet Mail Extensions)

1. Introduction

MIME is an extension of the original SMTP (Simple Mail Transfer Protocol). SMTP was originally designed to handle only 7-bit ASCII text, which meant it could not send images, audio, video, or even non-English characters. MIME was developed to allow the exchange of different types of data files over standard SMTP.

2. Why MIME is Needed

- **Original SMTP Limitations:** It cannot transfer binary files (executable programs) or multimedia (JPG, MP3).
- **MIME Solution**: It transforms non-ASCII data into 7-bit ASCII data at the sender's side and reverses it at the receiver's side.

3. MIME Header Elements

To support multimedia, MIME adds five new headers to the original SMTP message:

1. **MIME-Version**: Indicates the version of MIME being used (usually 1.0).
2. **Content**-Type: Describes the type of data (e.g., `image/jpeg` or `text/html`).
3. **Content-Transfer**-Encoding: Specifies how the data was wrapped into ASCII (e.g., Base64 or Quoted-printable).
4. **Content-ID**: Uniquely identifies the content in a message.
5. **Content-Description**: A text description of the body (e.g., "This is a photo of the campus").

4. Content Types and Subtypes

MIME defines a standardized set of types:

- **Text: `plain`, `html`**
- **Image: `jpeg`, `gif`, `png`**
- **Audio/Video: `basic`, `mpeg`, `mp4`**
- **Application: `pdf`, `zip`, `octet-stream` (for binary files)**
- **Multipart: Used when an email has multiple parts, like text and an attachment.**

DHCP (Dynamic Host Configuration Protocol)

1. Introduction

DHCP is a network management protocol used to automatically assign IP addresses and other communication parameters to devices on a network.

- **Model**: Client-Server Paradigm.
- **Transport Layer**: Uses UDP (Port 67 for Server, Port 68 for Client).

2. How DHCP Works (The DORA Process)

This is the most important part of a 10-mark answer. The process consists of four steps:

1. **Discovery** (DHCPDISCOVER): The client broadcasts a message to find a DHCP server.
2. **Offer** (DHCPOFFER): The server responds with an available IP address, subnet mask, and lease duration.
3. **Request** (DHCPREQUEST): The client "asks" to use the offered IP address.
4. **Acknowledgment** (DHCPACK): The server confirms the assignment and the client starts using the IP.

3. Key Parameters Assigned by DHCP

- **IP Address**
- **Subnet Mask (e.g., 255.255.255.0)**
- **Default Gateway (The router's IP)**
- **DNS Server IP**
- **Lease Time (How long the device can keep the IP)**

4. Advantages

- **Automation**: No need for manual configuration of every PC or phone.
- **Efficiency**: Prevents IP address conflicts (two devices having the same IP).
- **Mobility**: A laptop can move from one network to another and get a new IP automatically.

TELNET (Teletype Network)

1. Introduction

TELNET is one of the oldest Application Layer protocols. It provides a Remote Login facility, allowing a user on one computer to log into another computer on the network and execute commands as if they were sitting right in front of it.

- **Model: Client-Server.**
- **Transport Layer: Uses TCP on Port 23.**

2. Concept of Network Virtual Terminal (NVT)

Because different computers use different character sets (like ASCII vs. EBCDIC), TELNET uses an NVT.

- **The Telnet Client translates local characters into NVT characters.**
- **The Telnet Server translates NVT characters into the format the remote host understands.**

3. Working Mechanism

1. **The user starts the Telnet client and connects to the server's IP/Port 23.**
2. **The client establishes a TCP connection.**
3. **The user enters a username and password.**
4. **Once logged in, the user enters commands, which are sent to the server. The server executes them and sends the output back to the client's screen.**

4. TELNET Options and Commands

TELNET is highly negotiable. It uses specific codes for commands:

- **WILL/WONT**: Used by a party to offer/refuse to perform an option.
- **DO**/DONT: Used to request that the other party perform/stop performing an option.
- **Example** Options: Echoing characters, changing terminal type, or suppressing "Go Ahead."

5. Security Concern (Crucial Point)

TELNET is insecure. It sends all data—including passwords—in plain text. Anyone "sniffing" the network can see your credentials. In modern networks, it has been largely replaced by SSH (Secure Shell).